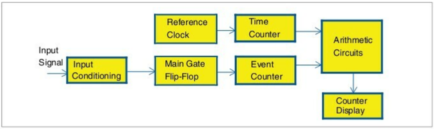

Fig. 5: Block diagram of a general frequency meter with reciprocal counting

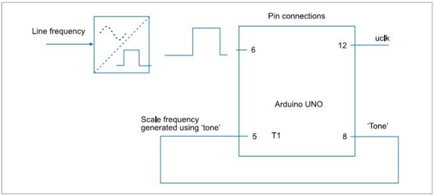

Fig. 6: Simplified block diagram of the line-frequency meter

Principle of operation

A generalised frequency meter with reciprocal counting is shown in Fig. 5.

An exact number of input cycles are counted in event counter and, simultaneously, the reference clock is counted in time counter for the same duration.

An equation can be written as:

Ns x Ts = Nr x Tr

where Ns is the number of cycles of the input frequency (50 counts in this case) and Ts is the period of the input signal. Nr is the number of cycles of the reference frequency counted within this time and Tr is the period of the reference clock (0.2 milliseconds or 0.2ms in this case).

Frequency of signal is given by:

Fs = 1/Ts = Ns/(Nr x Tr)

= 50/(Nr x 0.2ms) = 250000/Nr

If line frequency is exactly 50Hz, this equation will give:

Fs = (250000/5000) = 50

That is, to obtain a resolution of 0.01Hz, value in the display will be 50.00.

Arduino Uno is an excellent platform for the measurement of line frequency using the reciprocal counting method. A simplified block diagram of the line-frequency meter is shown in Fig. 6.

In this figure, line-voltage sine-wave is converted into a square wave of TTL levels using only a transistor. This is sufficient because the event counter counts only integral cycles and that eliminates zero-crossing errors. This square wave is applied to port pin 6 of Arduino Uno.

Square-wave cycles are counted by the software and the event counter. Clock frequency for the time counter is obtained from Arduino itself by the use of Tone() function in the Arduino code as explained in software section. This signal is applied to counter T1.

1. Event counter software counts exactly 50 cycles of the input signal.
2. Time counter simultaneously counts the number of reference pulses.
3. A simple reciprocal calculation then finds the frequency.

A counter similar to the circuits based on ICL7217 or NS74C926 can